0.1 ETCS

ETCS is a component of the broader European Rail Traffic Management System (ERTMS) and represents a significant advancement in train control technologies. It is designed to replace the numerous incompatible safety systems previously used by European railways, allowing cross-border interoperability and improving the safety and efficiency of railways. ETCS standardizes train control and command systems and includes continuous train monitoring via a digital radio-based signalling system. ETCS, much like FAO and GoA separates itself into five levels, these definitions are pulled from the latest update to ETCS, Control Command and Signalling (CCS) Interoperability (TSI) 2023. Specifically, the transport Europa document "ETCS Levels and Modes", Alstrom's document "ETCS, DIFFERENT LEVELS TO MATCH CUSTOMER'S NEEDS" and finally, the legislation documentat "Control Command and Signalling TSI, Commission Implementing Regulation (EU) 2023/1695 of 10 August 2023"[1] [2]

- Level 0 Trains with ETCS equipment run on lines without ETCS infrastructure. It's the most basic level, essentially serving as a compatibility mode for trains equipped with ETCS to operate on non-equipped tracks.
- Level National Train Control Refers to trains equipped with ETCS operating on lines equipped with legacy national systems. This level allows for a transitional phase where older systems are still in use but under the supervision of ETCS.
- Level 1 An add-on system where ETCS works in conjunction with existing lineside signals and train detectors. It involves non-continuous communication between the train and trackside through Eurobalises [3], enhancing safety and interoperability without complete infrastructure overhaul.
- Level 2 Enhances Level 1 by introducing continuous data communication between the train and a Radio Block Centre (RBC) via GSM-R radio, allowing for the optional removal of lineside signals. This level significantly increases line capacity and operational efficiency. This version of Level 2 is considered legacy however since a 2023 reclassification of the ETCS levels.
- Combined Level 2 and 3 (CL2/3) Originally, levels 2 and 3 were separate, however, now, level 2 has been classified as Level 2 (legacy) and Combined Levels 2 and 3. This development of ETCS has removed both the need for trackside detection and also has moved more of the technology on board. ETCS CL2/3 is separated into two subclassifications, ETCS CL2/3 Full Supervision (trackside assistant classified as limited supervision) and CL2/3 Automatic Driving (where ATO conditions are fulfilled.)

ETCS operates on a multi-level system similar to the GoA or FAO system, however, these levels are not without their own complications. Recent developments and documentation

[1] **https://transport.ec.europa.eu/transport-modes/rail/ertms/what-ertms-and-how-does-it-work/etcs-level**
https://www.era.europa.eu/domains/technical-specifications-interoperability/control-command-and-signallin
https://www.ertms.net/wp-content/uploads/2024/03/ERTMSETCS-Levels-Updated-2024-Edition.pdf.

[2] Thank you once again to our friend over at Leeds law school Mia, a real life saver on this one.

[3] Special communication device attached to rails for the purpose of ETCS.

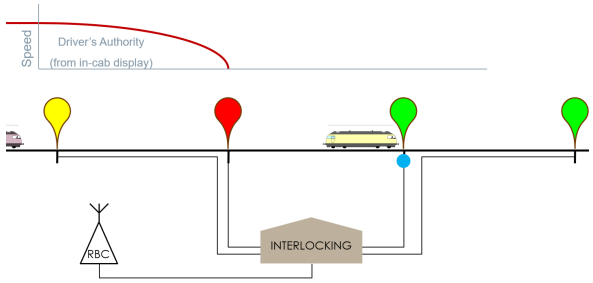

Figure 1: ETCS Level 2: part 1

from official European bodies have introduced conflicting interpretations. The CCS Technical Specification for TSI 2023 merged ETCS Level 3 into Level 2 as a variant termed "Level 2 Fixed Virtual Block/Moving Block." The Commission Implementing Regulation (EU) 2023/1695 of 10 August 2023 disregards FVB's and level 3 entirely, however discusses that systems may exist using older baselines so long as they meet the mandatory ERTMS specifications.

According to Commission Implementing Regulation (EU) 2023/1695 of 10 August 2023 Document 32023R1695[4] classification of ETCS requires meeting minimum requirements and gaining approval from member states for formal assessment within the specific subsystem. Furthermore, ETCS exists within the bounds of ERTMS, however, a full exploration of the ERTMS framework is slightly outside the bounds of this thesis.

0.2 Post Thesis Collaboration

Towards the tail-end of writing this document, we have began collaboration with Govia Thameslink Railway regarding the implementation of the SACRED methodology to assess ATO capabilities on the London Thameslink, as part of this collaboration, we have been given the Class 700 ETCS/ATO training material, as such, we have been given permission to share some further details regarding the specific operation of ETCS within the United Kingdom.

In summary, according to the legacy specification for ETCS Level 2, systems operate on a block-based architecture, as a train passes a ballast, it passes information to an interlocking centre, which passes the information to a Remote Broadcasting Centre, the interlocking centre ensures that the previous block has been cleared and the next block is now occupied, this updates signals through the RBC and shifts the drivers 'movement authority.' As demonstrated in figures 1 and 2

[4]European Union. *Commission Implementing Regulation (EU) 2023/1695 of 10 August 2023 on the technical specification for interoperability relating to the control-command and signalling subsystems of the rail system in the European Union and repealing Regulation (EU) 2016/919 (Text with EEA relevance)*. English. Aug. 2023. URL: `https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32023R1695&qid=1694158367331`.

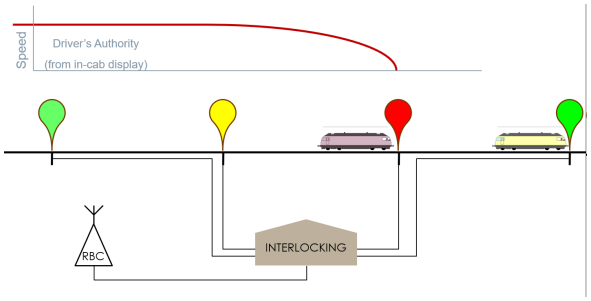

Figure 2: ETCS Level 2: part 2